

Troubleshooting Network Connectivity — Guided Notes

ANSWER KEY

Unit 5.2.2 | CompTIA Network+ N10-009 Obj. 5.2 | N10-008 5.2

For instructor use / distribute after students complete the notes.

Question 1

A workstation has IP address 169.254.45.3. This tells you the client _____ to contact a DHCP server. The client self-assigned an _____ address. From this state, the client can communicate with _____ but nothing else.

Answer: Failed; APIPA (Automatic Private IP Addressing / link-local). Other APIPA hosts on the same local segment.

Question 2

Two devices have the same IP address on the same VLAN. Describe the symptom a user would experience and name the command on the affected client that might reveal a changing MAC address for the gateway.

Answer: Intermittent connectivity — sessions work for moments then drop as ARP responses alternate between the two devices' MACs. Command: `arp -a` (the gateway's MAC entry flips between two different values).

Question 3

Complete the four-step diagnostic sequence for 'I can't get to the internet.'

Step 1: `ping \_\_\_\_\_` — tests local Layer 3 connectivity.

Step 2: `ping \_\_\_\_\_` — tests internet path by IP, bypassing DNS.

Step 3: `nslookup \_\_\_\_\_` — tests DNS resolution.

Step 4: If Step 2 works but Step 3 fails, the problem is _____.

Answer: Step 1: ping . Step 2: ping 8.8.8.8 (or any public IP). Step 3: nslookup google.com (any public domain). Step 4: DNS failure.

Question 4

A client can ping the gateway and external IPs but cannot reach port 443 on an internal server.

`Test-NetConnection 10.1.50.10 -Port 443` returns a timeout. What are two possible causes?

Answer: A network firewall rule is blocking TCP 443 to that server; or the Windows Firewall on the destination server is blocking inbound 443; or the service (web server) is not running on that host.

Question 5

A VLAN that exists on an access switch but is not added to the _____ between that switch and the distribution layer will have Layer 2 connectivity locally but cannot reach the _____. Clients will appear to have valid IPs but go nowhere.

Answer: Trunk (allowed VLAN list); default gateway / rest of the network.

Question 6

NAT misconfiguration can cause internal hosts to send packets to the internet with _____ source addresses that are dropped by ISP routers. A new subnet that wasn't added to the NAT _____ list is the most common cause after a VLAN expansion.

Answer: Private (RFC 1918); access control list (ACL) / NAT pool.

Question 7

Multiple users on one switch cannot reach the internet, but users on other switches are fine. The uplink from that switch shows _____ in `show interface`. What is the most efficient next step to narrow the cause?

Answer: Err-disabled (or down/down); investigate why the uplink err-disabled — check switch logs for the error type (BPDU guard, port security violation, loop detection) before re-enabling.

Question 8 — Real World Application

REAL WORLD: A teacher moves to a new classroom. Their laptop has the correct IP, gateway, and DNS. They can ping the gateway. They can ping 8.8.8.8. But no websites load. Nslookup for any public domain returns the correct IP. Describe what you would check next and why.

Answer: The DNS and network path are working. The next check is a firewall rule — try `Test-NetConnection google.com -Port 443` and port 80. If both time out, a network firewall is likely blocking outbound HTTP/HTTPS from the new classroom's port or VLAN. Check whether the port is in the correct VLAN and whether the VLAN has outbound permit rules for ports 80 and 443.